RESEARCH PAPER HUB

Week 4 Check-in — Milestone 2

Project Plan, Journey Maps, Measurable Objectives & HAI Principle Ladder

Track: Research Paper Discovery Hub
Principle: Transparency

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Project Plan v1

Week	Main Activity	Expected Output
1–2	Problem & user research	Problem definition + target users
3	Concept generation	Concept A + Concept B
4	Objectives & HAI ladder	Milestone 2
5–6	Knowledge source preparation	Curated paper collection + metadata
7–8	Retrieval & grounding	Initial retrieval pipeline
9	Citation checking	Citation validation prototype
10	Claim labeling	Evidence/claim classification
11	Research workflow	6-stage interaction flow prototype
12	Show Evidence	Evidence inspection feature
13	User testing & evaluation	Metrics + usability feedback
14	Refinement & final presentation	Final prototype + evaluation

Journey Map — Beginner Research Student

Journey Stage	What the Student Does	Pain Point	How Transparency Helps
Start	Starts with a general research interest and asks the AI for guidance.	Does not know where to start or how to narrow the topic.	The AI clearly explains what it is, what sources it uses, and its limitations.
Ask	Describes the topic and asks for help formulating a research question.	May not know whether the suggested question is suitable or AI-generated.	Suggestions are clearly labeled as **AI recommendations**.
Understand the Answer	Reviews suggested papers and reads explanations of their methods, findings, and limitations.	Cannot easily distinguish information supported by papers from AI interpretation.	Claims receive evidence-status labels, citations, and a **Show Evidence** option.
Act / Fix a Problem	Selects relevant papers, refines the research question, and checks unclear or conflicting evidence.	May trust unsupported claims or become confused by conflicting findings.	The system flags weak or conflicting evidence and allows the student to inspect the original passages.
Finish	Reviews the collected evidence and produces a preliminary research brief.	May finish with conclusions that have not been properly verified.	The final brief keeps source IDs, separates verified information from interpretations, and includes a verification checklist.

Measurable Objectives

Objective	Metric	Target
Improve the quality of AI-generated research claims.	Citation Support Rate	≥ 90%
Help users understand whether a claim is supported by evidence or generated as an AI interpretation/recommendation.	Claim Classification Accuracy	≥ 90%
Make the research process easy for students to complete.	Task Completion Rate	≥ 80%
Help users successfully verify important claims using the available evidence.	Evidence Verification Success Rate	≥ 75%

HAI Principle Ladder v1

Level	How Transparency is Applied	Measurable Objective
1. Inform	The system informs users about its capabilities, knowledge sources, and limitations before they begin the research process.	≥ 90% of users can correctly identify the system's main source and limitations.
2. Explain	The system explains where research claims come from by providing claim labels, citations, and supporting evidence when available.	≥ 85% of users can correctly distinguish source-supported claims from AI interpretations and recommendations.
3. Allow Control	Users can approve, edit, or reject research questions, selected papers, and potential research gaps.	≥ 80% of users can complete the main workflow while making the required decisions.
4. Enable Feedback	Users can provide feedback by marking evidence as relevant, uncertain, or irrelevant and can flag problematic claims.	≥ 75% of users can successfully provide and apply feedback to the research workflow.

Tool Choices

Component	Selected Tool / Approach	Purpose
AI Model	Hosted LLM API — model/provider to be finalized	Generate explanations, research-question suggestions, paper analysis, comparisons, and research-brief drafts.
Backend	ASP.NET Core Web API	Manage users, research sessions, papers, claims, evidence, citations, and communication with the AI model.
Interface	Web-based interface	Allow students to interact with the research workflow, review claims, inspect evidence, and provide feedback.
Knowledge Source	Curated collection of academic papers	Provide the trusted knowledge base used to ground research-related answers.
Grounding	Retrieval-Augmented Generation (RAG)	Retrieve relevant passages from the curated papers and provide them to the AI model before generating research-related answers.
Evidence & Citation	Retrieved passages + citation validation	Connect claims to their supporting sources and identify unsupported or insufficiently supported claims.

How Answers Will Be Grounded

The system will not rely only on the AI model's internal knowledge for research-related answers.

Instead, relevant academic papers will be retrieved from the curated knowledge source. The retrieved passages will then be provided to the AI model as evidence for generating the response.

The generated claims will be checked against the available evidence and assigned an appropriate status, such as **Source-supported, AI interpretation, AI recommendation, or Unverified hypothesis**.

When evidence is available, the **Show Evidence** feature will allow the student to inspect the supporting passage and source information.

Appendix A — Prompt Log

Tool	Purpose	Prompt / Task	Output	Team Decision
ChatGPT	Measurable Objectives	Asked for measurable objectives with target numbers, including an AI-quality metric, a user-experience metric, and a Transparency-related metric.	Suggested metrics for citation support, claim classification, task completion, and evidence verification.	Selected four measurable metrics and defined target values for each.
Gemini	Journey Mapping	Asked to create a journey map for the beginner research student covering Start, Ask, Understand the Answer, Act/Fix a Problem, and Finish.	Generated the student's actions, pain points, and possible Transparency support at each stage.	Reviewed the journey and aligned it with the Research Paper Hub workflow.
Claude	HAI Principle Ladder	Asked to map Inform, Explain, Allow Control, and Enable Feedback to the Research Paper Hub's Transparency principle.	Proposed a four-level HAI Principle Ladder with measurable objectives.	Reviewed the mapping and connected each level to a measurable target.
Perplexity	Tool & Grounding Research	Asked to explore suitable approaches for selecting an AI model, backend/interface tools, and grounding research answers in academic sources.	Provided possible technology and grounding approaches for the project.	Used the findings as input for the proposed Tool Choices and kept final technology decisions open where not yet finalized.